

DAA

Case Assessment Tool - I

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Analytical problem solving

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Course Code: CSA0613

3B. Linear search - duplicate target elements (Analytical problem solving)

1. problem understanding

Given the unsorted dataset:-

[9, 42, 11, 42, 56, 73, 42]

find the first occurrence of the target element 42 using linear search

Requirements:-

- Explain the search step-by-step
- Count the number of comparisons until the first match.
- Analyze best-case, average-case, and worst-case time complexity.
- Verify the correctness
- Interpret the efficiency when duplicate target values are present.

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2. method selection

Linear search is selected because:-

- The data is unsorted
- It checks each element from left to right until the target is found.
- It returns the first occurrence of the target

3. Step-by-step solution:-

Dataset:-

Index	Value:-
0	19
1	42
2	11
3	42
4	56
5	73
6	42

target = 42

Step	element Compared	Comparison	Result:-
1	19	$19 = 42?$	No
2	42	$42 = 42?$	Yes (found)

The first occurrence has been found:-

* Calculations:-
→ first occurrence
→ Number of
→ final answer
→ target
→

4. Calculations:-

- first occurrence index = 1
- Number of comparisons = 2

5. Final Answer:-

- Target element: 42
- first occurrence: index 1
- comparisons required: 2

6. Time Complexity Analysis

→ Best case: $O(1)$

Target is the first element.

→ Average case: $O(n)$

Target is found around the middle of the list

→ Worst case: $O(n)$

Target is the last element or not present

7. Verification of correctness:-

The dataset contains 42 at indices 1, 3, and

6.

Linear search scans from left to right and stops at the first match, which is at index 1.

The result is correct

8. Interpretation of Results:-

* linear search successfully finds the first occurrence of duplicate values:-

* duplicate elements appearing later are not checked after the first match is found.

* In this example, only 2 comparisons were needed, making the search efficient.

* If the first occurrence were farther on the list, more comparisons would be required, so the algorithm's performance still depends on the position of the first matching elements:-